

VIDHURA POLGOLLA

Mechanical & Manufacturing Engineering Graduate | Automation & Manufacturing

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PROFILE

Mechanical and Manufacturing Engineering graduate at the University of Ruhuna, with industrial experience in automation, machine design and manufacturing improvement across FMCG environments. Skilled in PLC/HMI programming, control system integration, SolidWorks, FEA and electro-pneumatic systems, with hands-on experience delivering measurable improvements in cost, changeover time and process accuracy.

EDUCATION

BSc.Eng (Hons) in Mechanical and Manufacturing Engineering

2026

Faculty of Engineering, University of Ruhuna

St. Sylvester's College, Kandy

2015 - 2020

- G.C.E. Advanced Level 2020 - Physics (A), Chemistry (A), Combined Mathematics (B); Ordinary Level 2015 - 9 A's

WORK EXPERIENCE

Intern – Automation Engineer

Aug 2025 - Nov 2025

Hemas Manufacturing (Pvt) Ltd, Dankotuwa

- Delivered automation and continuous-improvement projects on live FMCG production lines, targeting machine downtime, operator intervention and process efficiency.
- Designed 3D concepts for a proposed automated shampoo filling machine in SolidWorks.
- Developed and modified PLC/HMI control systems across Mitsubishi, Siemens and Allen-Bradley platforms.
- Executed electrical control panel wiring and instrumentation integration for production equipment.
- Applied TPM, RBM, CBM, Lean, TQM and 5S practices to strengthen equipment reliability and manufacturing performance.
- Collaborated with engineering, production and quality teams to troubleshoot equipment and implement practical manufacturing solutions.

Mechanical Engineering Trainee

May 2024 - Aug 2024

Ceylon Biscuits Limited, Pannipitiya

- Maintained and repaired conveyors, gearboxes, boilers and packaging machines in a large-scale food manufacturing plant.
- Diagnosed pneumatic and electromechanical faults on live production lines to restore output with minimal stoppage.
- Designed a vibrating spice sprinkling machine and a packet air-trapping mechanism for vertical form-fill-seal machines.
- Managed maintenance planning, industrial documentation and inventory control in SAP ERP.

TECHNICAL SKILLS

Design & CAD: SolidWorks · AutoCAD · SolidCAM · Onshape · 3D CAD Assembly · Machine Design

Analysis & Simulation: ANSYS (FEA) · MATLAB / Simulink · JMAG Designer · Festo FluidSim · Motion and Kinematic Analysis

Automation & Controls: PLC Programming - Mitsubishi GX Works, Allen-Bradley, Siemens, Haiwell, IDEC, Delta · HMI/SCADA Design · Sensors & Instrumentation · Pneumatics · Servo Drives · VFDs · Industrial Robotics

Electrical & Hardware: Control Panel Wiring · Microcontroller-Based Control Hardware · PCB Design

Manufacturing & Systems: TPM · CBM · Preventive & Predictive Maintenance · Lean · Kaizen · 5S · Six Sigma (Yellow Belt) · Root Cause Analysis · GMP · ISO Standards · Lathe Operation, Grinding and Arc Welding

Productivity: Microsoft Office · MS Project · SAP ERP

KEY PROJECTS

Automated Glove Pin-Hole Detection Machine

Aug 2026

Capstone Project, with Lalan Rubbers

Designed and fabricated a modular embedded automation system for pressure-decay micro-leak testing of industrial gloves, integrating STM32-based control, pressure sensing, HMI, Modbus communication and pneumatic actuation.

- Developed the central control logic for automated inflation, settling and pressure-decay testing, with configurable test parameters and real-time pressure monitoring through an HMI.
- Integrated high precision sensors with a 16-bit ADC for pressure measurement and leak detection.
- Designed a custom PCB breakout/interface board and isolation circuits for robust integration between control electronics, sensors, pneumatic actuators and machine indicators.
- Reduced control-system cost by approximately 75%, from an industry-standard ~LKR 400,000 to ~LKR 100,000, while validating the Phase 1 system as a modular platform for future machine-vision inspection.

Tools: STM32 Microcontroller · I2C Communication · Honeywell 24PC Pressure Sensors · ADS1115 16-bit ADC · Custom HMI Panel

Double Feeding Corrugated Box Forming Machine for Mixed-Model Lines

Aug 2026

Final Year Design Project, with Hemas Consumer Brands

Designing a high-speed automated case erector to eliminate a production bottleneck caused by a single 6-axis robot serving two packing lines at a 25-second cycle time.

- Developing a fixed datum line architecture in which only one conveyor moves during changeover, sharply reducing mechanical complexity.
- Building the full 3D CAD assembly and running FEA on the main frame, gantry and load-bearing components.
- Conducting motion and kinematic studies to verify servo strokes, pneumatic timing and clash-free flap folding, and sizing AC servos, pneumatic cylinders and venturi vacuum ejectors by calculation.
- Targeting a cycle under 10 seconds, two preset box sizes up to 600 x 350 x 300 mm from a dual magazine, a 3.0 x 2.5 m footprint and a budget under LKR 5 million.

Tools: SolidWorks · Kinematic Motion Analysis · 24 V DC PLC · AC Servo Drives · MJ50 Overhead Gantry · Venturi Vacuum Ejector

Automation of LABSA Pumping System using PLC and HMI

Sep 2025

Hemas Manufacturing (Pvt) Ltd, Dankotuwa, Kaizen Project

Replaced manual, sight-mark measurement of Linear Alkylbenzene Sulfonic Acid in the DIVA detergent line with automated, flow-based dosing.

- Developed PLC control logic for automated pump operation and configured an HMI for real-time monitoring and operator control.
- Integrated flow meter pulse output for accurate dosing, implemented safety interlocks and emergency stop, and carried out panel wiring, labelling and system integration.
- Achieved over 99% dosing accuracy, reduced the scrap factor from 1.2 to 0.8 and generated an estimated LKR 1.2 million in annual savings with batch-wise traceability.
- Selected as the highest-scoring project in the Engineering Department Kaizen Team Competition.

Tools: Mitsubishi FX5U-32M PLC · GOT1000 HMI · GX Works2 · GT Designer · Endress+Hauser Promass F 500 Flow Meter

Changeover Time Reduction in a Volumetric Filling Machine

Aug 2025

Hemas Manufacturing (Pvt) Ltd, Dankotuwa

Eliminated manual reed switch repositioning on a shampoo filling line running at around 120 bottles per minute.

- Designed and implemented a bypass switching system that toggles between the adjustable reed switch and a fixed end-position switch, giving instant stroke changeover with no change to the PLC program.
- Reorganised and tagged wiring to cut wrench time and improve operator access.
- Reduced changeover time by 6 minutes per product change, lifting machine availability on a high-speed line.

Other Academic Projects

2024

- Zero-Power Magnetic Levitation Control System - nonlinear control system levitating a steel ball at zero coil current; modelled in SolidWorks, analysed in JMAG Designer and controlled with a PD+I controller in Simulink.
- Fully Automated Chinese Rolls Making Machine - conceptual mass-production system covering filling, folding, rolling, frying and batching, with an electro-pneumatic control circuit in Festo FluidSim and logic in Haiwell Happy PLC software.

AWARDS & ACHIEVEMENTS

- CAD Vortex SOLIDWORKS Competition 2026 - Champions
- Auto Mech PLC Competition 2026 - Champions
- Lean Six Sigma Yellow Belt Certification - MF Treinamentos
- Certificate Course in Career Guidance - University of Ruhuna
- Highest-scoring project, Engineering Department Kaizen Team Competition, Hemas - 2025

LEADERSHIP & VOLUNTEERING

- REF Media, Ruhuna Engineering Faculty Media Unit - President 2026, Vice President 2025, Editor 2024
- Ruhuna Engineering Student Chapter of the IESL - Editor 2024
- Mehewara, Students' Union, Faculty of Engineering, University of Ruhuna - Student Volunteer
- Senior Prefect of St. Sylvester's College, Kandy - 2017/18

REFEREES

- **Dr. Krishan Wickramasinghe** - Senior Lecturer, Dept. of Mechanical and Manufacturing Engineering, University of Ruhuna
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- **Mr. Udaya Kumara** - Senior Engineer, Ceylon Biscuits Limited, High-Level Road, Makumbura, Pannipitiya
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